Function notation

1 The point $(7, -6)$ satisfies the function $f(x)$. Write this using function notation.

2 If $f(x) = 9x^2 + 7x - 4$, find:

a $f(-4)$

b $f(10)$

3 If $f(x) = -6x + 4$, find:

a $f(4)$

b $f(0)$

4 If $f(x) = 4x + 4$, find:

a $f(2)$

b $f(-5)$

5 If $f(x) = 3x - 1$, find:

a $f(3)$

b $f(-4)$

6 If $g(x) = \dfrac{7x}{4}$, find:

a $g(5)$

b $g(-4)$

7 Consider the function $f(x) = 4 + x^3$.

a Evaluate $f(4)$

b Evaluate $f(-2)$

8 Consider the function $f(x) = 2x^2 - 2x + 5$. Evaluate $f\left(\dfrac{1}{2}\right)$.

9 Consider the function $f(x) = \sqrt{5x + 9}$. Find the exact value of:

a $f(0)$

b $f(2)$

c $f(-1)$

10 If $f(t) = \dfrac{t^3 + 27}{t^2 + 9}$, evaluate the following:

a $f(-3)$

b $f(3)$

c $f(4)$

11 Consider the function $f(x) = x^2 + 8x$. Write an expression for:

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12 Consider the function $f(x) = 2x^3 + 3x^2 - 4$.

a Evaluate $f(0)$.

b Evaluate $f\left(\frac{1}{4}\right)$.

13 If $j(x) = 3^x - 3^{-x}$, find the following, rounding your answers to two decimal places if necessary:

a $j(0)$

b $j(1)$

c $j(4)$

14 If $m(x) = \sqrt{12^2 - x^2}$, find:

a $m(4)$

b $m(0)$

c $m(9)$

d $m\left(\sqrt{2}\right)$

15 Consider the function $p(x) = x^2 + 8$.

a Evaluate $p(2)$.

b Form an expression for $p(m)$.

16 Consider the equation $x + 3y = 6$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$ for y.

c Find the value of $f(3)$.

17 Consider the equation $x - 4y = 8$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$.

c Find the value of $f(12)$.

18 Consider the equation $y + 6x^2 = 3 - x$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$.

c Find the value of $f(3)$.

19 Consider the equation $y - 4x^2 = 5 + x$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$.

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20 Consider the equation $-6x+5y=7$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$.

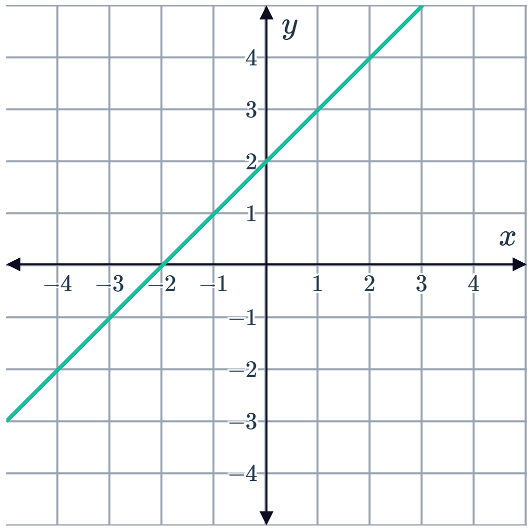
c Find the value of $f(3)$.

21 Use the graph of the function $f(x)$ to find each of the following values:

a $f(0)$

b $f(-2)$

c The value of x such that $f(x)=3$



22 Consider the relation $h(x)=-x^2+6x-6$.

a For any input value of x, state the maximum number of distinct output values $h(x)$ can produce.

b Is $h(x)$ a function? Explain your answer.

23 For each of the following pairs of variables, determine which is the independent variable and which is the dependent variable:

a Cost of pizza and size of pizza.

b Mark achieved on a test and time spent studying.

c Duration of a loan and amound borrowed.

d Time spent exercising and physical fitness level.

24 Consider the function $f(x)=x^2-49$.

a Find:

i $f(1)$ **ii** $f(8)$ **iii** $f(0)$

b If $f(x)=12$, what are the possible values for x, rounded to two decimal places?

25 Consider the function $g(x) = ax^3 - 3x +$ [redacted]

a Form an expression for $g(k)$.

b Form an expression for $g(-k)$.

c Is $g(k) = g(-k)$?

d Is $g(k) = -g(-k)$?

Algebra of functions

26 A function is defined as $f(x) = 2 + 3x - x^3$. Evaluate $f(-2) + f(4)$.

27 If $A(x) = x^2 + 1$ and $Q(x) = x^2 + 9x$, evaluate the following:

a $A(5)$

b $Q(4)$

c $A(3) + Q(2)$

28 Consider the function $f(x) = x^3 - 8x + 7$.

a Evaluate $f(4)$.

b Evaluate $f(-4)$.

c Form an expression for $f(a)$.

d Form an expression for $f(b)$.

e Form an expression for $f(a+b)$.

f Is $f(a) + f(b) = f(a+b)$ for all values of a and b?

29 Consider the function $f(x) = x^2 + 5x$. Find $f(a+h)$.

30 Consider the function $f(x) = x^2 - 2x$. Find $f(a+6)$.

31 Consider the function $f(x) = x^2 - 3x - 2$.

a Form an expression for $f(a-2)$.

b Form an expression for $f(a+h) - f(a)$.

32 Consider the function $f(x) = 4x^2 + 4x + 3$.

a Form an expression for $f(a+2)$.

b Form an expression for $f(a+h) - f(a)$.

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33 Consider the following table of values:

x	2	7	8	9
$f(x)$	4	14	16	18
$g(x)$	8	28	32	36

a Find $(f+g)(2)$.
b Find $(f+g)(7)$.
c Find $(f \times g)(2)$.
d Find $(g-f)(9)$.

34 Let $f(x) = x^2 - 1$ and $g(x) = 5x - 1$. Find $f(2) + g(2)$.

35 Let $f(x) = -5x + 3$ and $g(x) = x^2 - 7$.

a Find $(f \times g)(x)$.
b Evaluate $(f \times g)(-3)$.

36 If $f(x) = 3x - 5$ and $g(x) = 5x + 7$, find:

a $(f+g)(x)$
b $(f+g)(4)$
c $(f-g)(x)$
d $(f-g)(10)$

Applications

37 If $Z(y) = y^2 + 12y + 32$, find y when $Z(y) = -3$.

38 A function $f(x)$ is defined by $f(x) = (x+4)\left(x^2 - 4\right)$.

a Evaluate $f(6)$.
b Find all solutions for which $f(x) = 0$.

39 A graph of a quadratic equation of the form $y = ax^2 + bx + c$ passes through the points $(0, -7)$, $(-1, -8)$ and $(4, 77)$.

a Using the point $(0, -7)$, find the value of c.
b Substitute c and $(-1, -8)$ into the equation $y = ax^2 + bx + c$ to obtain an equation that describes a in terms of b.
c Substitute c and $(4, 77)$ into the equation $y = ax^2 + bx + c$ to obtain a simplified second equation in terms of a and b.
d Solve for a and b.

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40 The financial team at The Gamgee Coope[redacted] wants to calculate the profit, $P(x)$, generated by producing x units of wetsuits.

The revenue produced by the product is given by the equation is $R(x) = -\frac{x^2}{4} + 40x$. The cost of production is given by the equation $C(x) = 5x + 410$.
The profit is calculated as $P(x) = R(x) - C(x)$.

a Find an expression for $P(x)$ in terms of x.

b Find the values of the following:

i $R(70)$ **ii** $C(70)$ **iii** $P(70)$

c Sketch the graphs of $y = R(x)$, $y = C(x)$ and $y = P(x)$.

41 Elizabeth wants to calculate the cost to travel to Tehran and then Mexico City at certain times of the year. A program on her computer calculates $T(x)$, the total cost of this trip, by adding the cost of travelling to Tehran, $S(x)$, and the cost of travelling to Mexico City, $U(x)$. Based on historical data, the computer program uses the calculations:

$$S(x) = x^2 - 200x + 10\,227 \quad \text{and} \quad U(x) = 18x + 15\,423$$

where x is the number of days until the date of travel.

a Find an equation for $T(x)$.

b Find the values of the following:

i $S(91)$ **ii** $U(91)$ **iii** $T(91)$

c Sketch the graphs of $y = S(x)$, $y = U(x)$ and $y = T(x)$.

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